

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Update, Change]	138781	46.073276
1	[Git, Change, Create]	44929	14.915775
2	[Git, Change, Delete]	22781	7.562961
3	[Java, ByteCode, Method, Member]	13525	4.490104
4	[Java, ByteCode, Parameter]	13306	4.417399
5	[Git, Commit]	11390	3.781315
6	[File, Git]	8286	2.750832
7	[Java, ByteCode, Bound, ParameterizedType]	7286	2.418846
8	[Java, ByteCode, Bound]	7245	2.405235
9	[Git, Change, Rename]	7018	2.329874
10	[Java, ByteCode, Member, Field]	3606	1.197140
11	[Java, ByteCode, Bound, WildcardType]	2950	0.979357
12	[Java, Value, ByteCode, Annotation]	2936	0.974709
13	[Xml, Element]	2162	0.717753
14	[Java, ByteCode, Constructor, Method, Member]	2111	0.700821
15	[Xml, Text]	1450	0.481379
16	[Java, ByteCode, TypeVariable, Bound]	1111	0.368836
17	[Java, ByteCode, Method, Member, Lambda]	968	0.321362
18	[Type, File, Java, ByteCode, ResolvedDuplicate...	890	0.295467
19	[Type, File, Java, ByteCode, Class]	845	0.280528
20	[Json, Key]	739	0.245337
21	[Value, Json, Scalar]	723	0.240025
22	[Java, Value, ByteCode, Primitive]	673	0.223426
23	[Type, File, Java, ByteCode, JavaType]	649	0.215459
24	[Java, ByteCode, Method, Member, GenericDeclar...	578	0.191888
25	[Type, File, Java, ByteCode, ExternalType]	412	0.136778
26	[Git, Change, Copy]	298	0.098932
27	[Author, Git, Person]	295	0.097936
28	[Value, Array]	289	0.095944
29	[Committer, Git, Person]	242	0.080340

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

<Figure size 640x480 with 0 Axes>

Nodes per label combination (more than 0.5% overall)

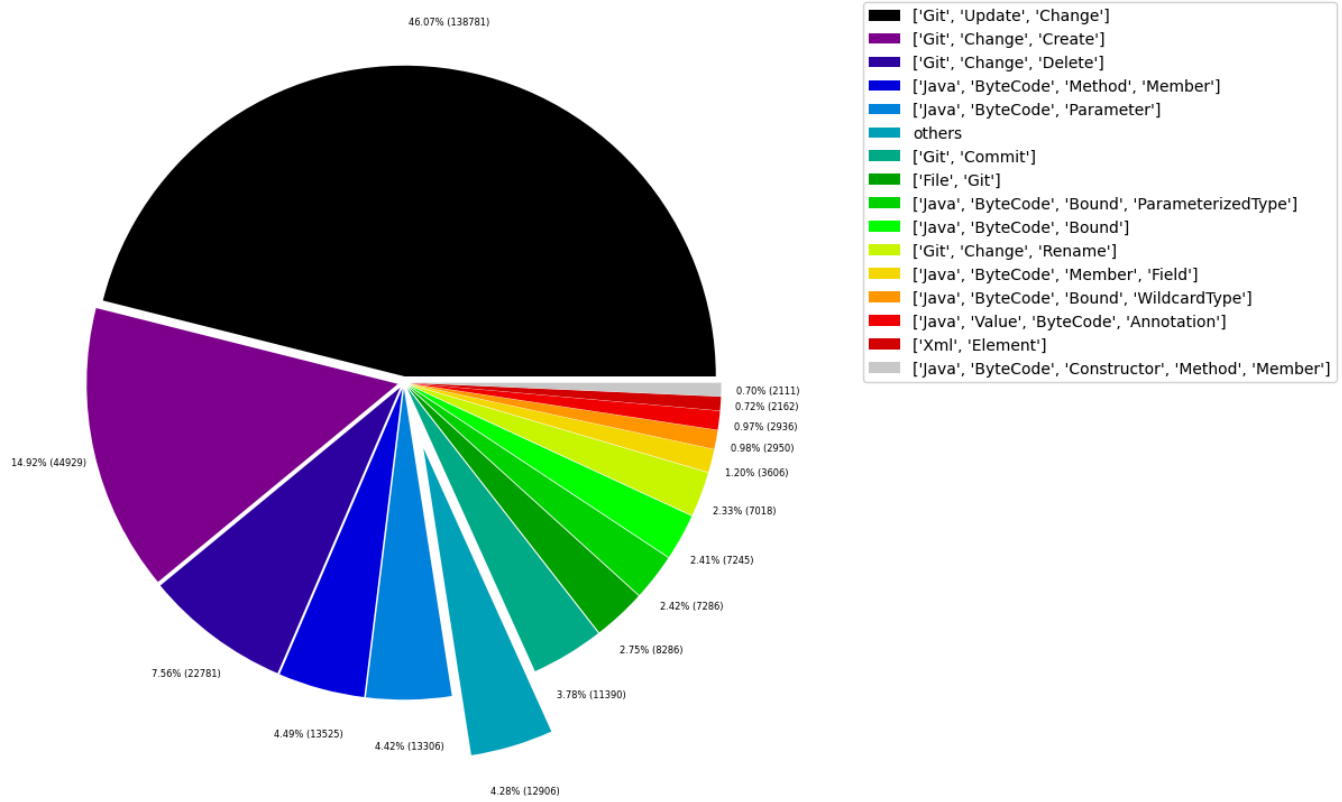


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, JQAssistant]	1	0.000332
1	[Git, Branch]	1	0.000332
2	[Repository, File, Git]	1	0.000332
3	[File, Json]	2	0.000664
4	[File]	3	0.000996
5	[Java, ByteCode, Constructor, Method, Member, ...]	4	0.001328
6	[Maven, Exclusion]	5	0.001660
7	[Value, Array, Json]	6	0.001992
8	[Type, File, Java, ByteCode, Void]	9	0.002988
9	[File, Maven, Xml, Pom, Document]	9	0.002988
10	[Java, ManifestSection]	9	0.002988
11	[File, Artifact, Jar, Archive, Zip, Java]	9	0.002988
12	[File, Java, Manifest]	9	0.002988
13	[File, Java, ServiceLoader]	10	0.003320
14	[File, Java, Properties]	12	0.003984
15	[Maven, ExecutionGoal]	16	0.005312
16	[Maven, PluginExecution]	16	0.005312
17	[Xml, Attribute]	18	0.005976
18	[JQAssistant, Rule, Concept]	19	0.006308
19	[Maven, Plugin]	21	0.006972
20	[Maven, Configuration]	21	0.006972
21	[Type, File, Java, ByteCode, Throwable, Extern...	21	0.006972
22	[Type, File, Java, ByteCode, Throwable, Resolv...	22	0.007304
23	[Type, File, Java, ByteCode, Enum]	28	0.009296
24	[Type, File, Java, ByteCode, PrimitiveType]	30	0.009960
25	[Xml, Namespace]	36	0.011951
26	[Type, File, Java, ByteCode, Annotation]	44	0.014607
27	[File, Directory]	44	0.014607
28	[Type, File, Java, ByteCode, Class, Throwable]	55	0.018259
29	[Type, File, Java, ByteCode, GenericDeclaratio...	85	0.028219

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Nodes per label combination (less than 0.5% overall)

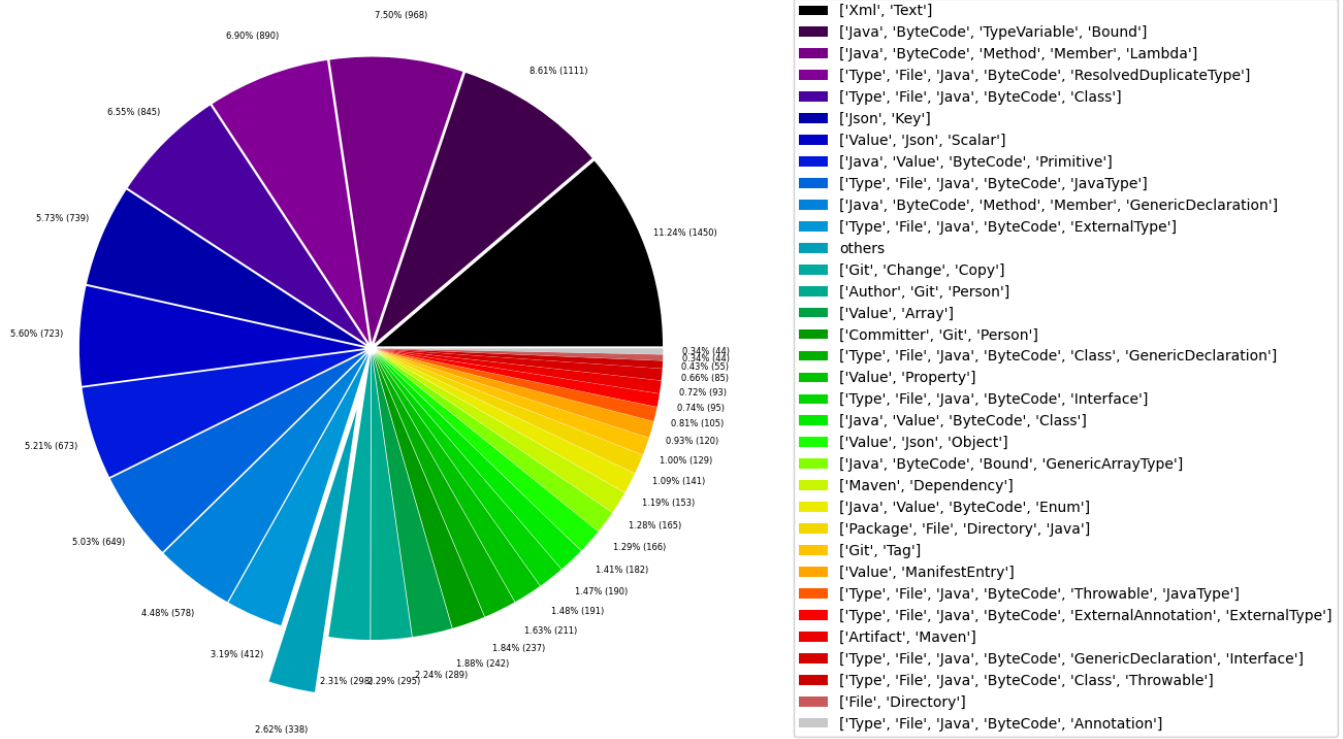


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

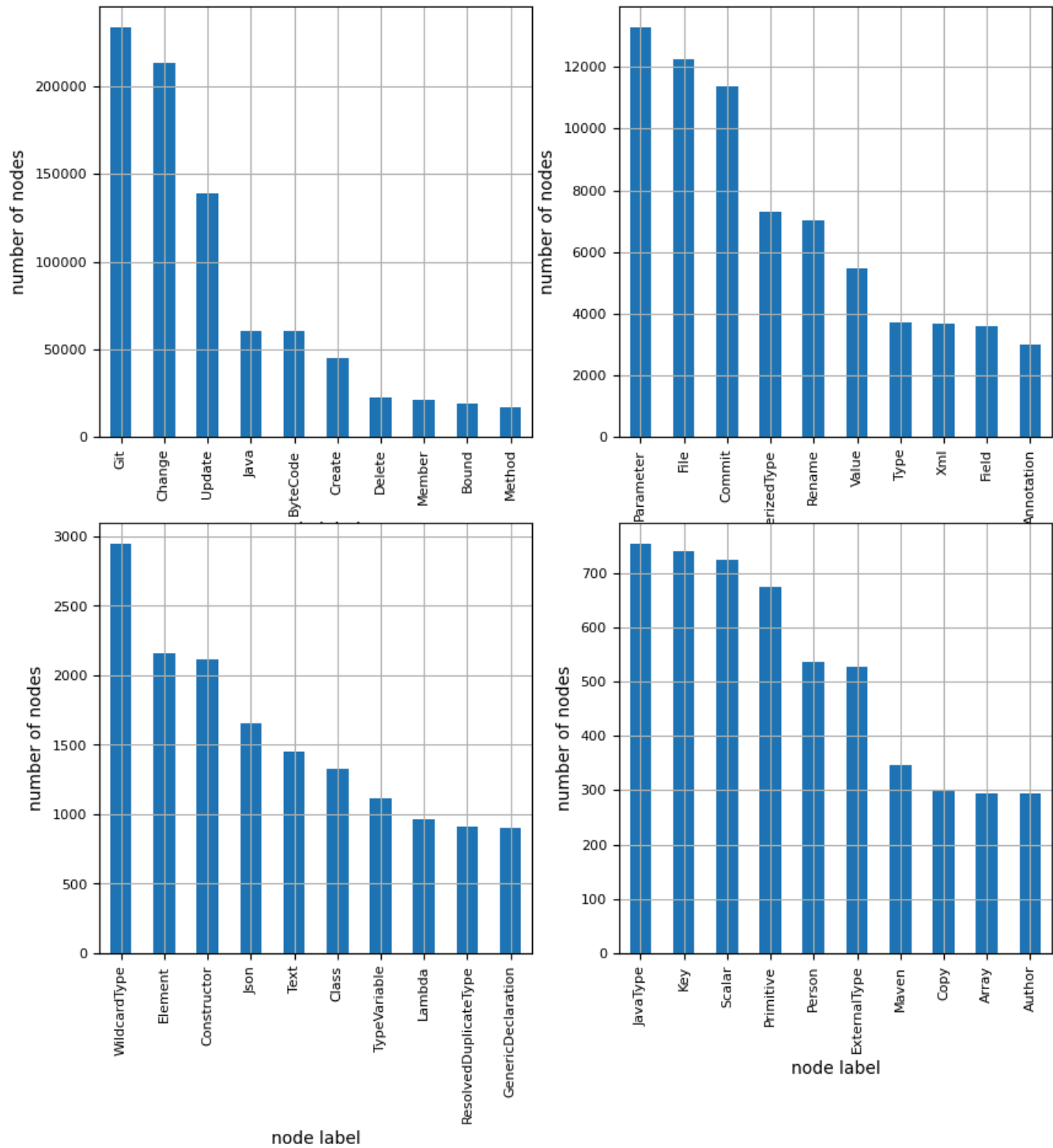
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	234151	77.734730
1	Change	213807	70.980818
2	Update	138781	46.073276
3	Java	60716	20.156830
4	ByteCode	60526	20.093753
5	Create	44929	14.915775
6	Delete	22781	7.562961
7	Member	20792	6.902642
8	Bound	18758	6.227383
9	Method	17186	5.705502
10	Parameter	13306	4.417399
11	File	12244	4.064830
12	Commit	11390	3.781315
13	ParameterizedType	7286	2.418846
14	Rename	7018	2.329874
15	Value	5483	1.820276
16	Type	3718	1.234322
17	Xml	3675	1.220047
18	Field	3606	1.197140
19	Annotation	2980	0.989317
20	WildcardType	2950	0.979357
21	Element	2162	0.717753
22	Constructor	2115	0.702149
23	Json	1652	0.548440
24	Text	1450	0.481379
25	Class	1327	0.440545
26	TypeVariable	1111	0.368836
27	Lambda	968	0.321362
28	ResolvedDuplicateType	912	0.302771
29	GenericDeclaration	904	0.300115
30	JavaType	754	0.250317
31	Key	739	0.245337
32	Scalar	723	0.240025
33	Primitive	673	0.223426
34	Person	537	0.178276
35	ExternalType	528	0.175288
36	Maven	346	0.114867
37	Copy	298	0.098932
38	Array	295	0.097936
39	Author	295	0.097936

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

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Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 936040

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	213807	22.841652
1	MODIFIES	213807	22.841652
2	UPDATES	138781	14.826396
3	CREATES	52245	5.581492
4	INVOKES	36692	3.919918
5	DELETES	29799	3.183518
6	COMMITTED	22780	2.433657
7	DEPENDS_ON	22504	2.404171
8	OF_TYPE	21945	2.344451
9	DECLARES	21252	2.270416
10	OF_RAW_TYPE	17367	1.855369
11	HAS	14444	1.543096
12	HAS_PARENT	13921	1.487223
13	RETURNS	12858	1.373659
14	HAS_COMMIT	11390	1.216828
15	READS	9385	1.002628
16	HAS_ACTUAL_TYPE_ARGUMENT	8414	0.898893
17	HAS_FILE	8286	0.885219
18	RENAMES	7018	0.749754
19	OF_GENERIC_TYPE	6000	0.640998
20	RESOLVES_TO	4245	0.453506
21	SIMILAR	4078	0.435665
22	WRITES	3948	0.421777
23	CONTAINS	3928	0.419640
24	HAS_NEW_NAME	3662	0.391223
25	RETURNS_GENERIC	3597	0.384278
26	CHANGED_TOGETHER_WITH	3034	0.324131
27	ANNOTATED_BY	2924	0.312380
28	REQUIRES	2233	0.238558
29	HAS_FIRST_CHILD	2162	0.230973

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

<Figure size 640x480 with 0 Axes>

Relationship types (more than 0.5% overall)

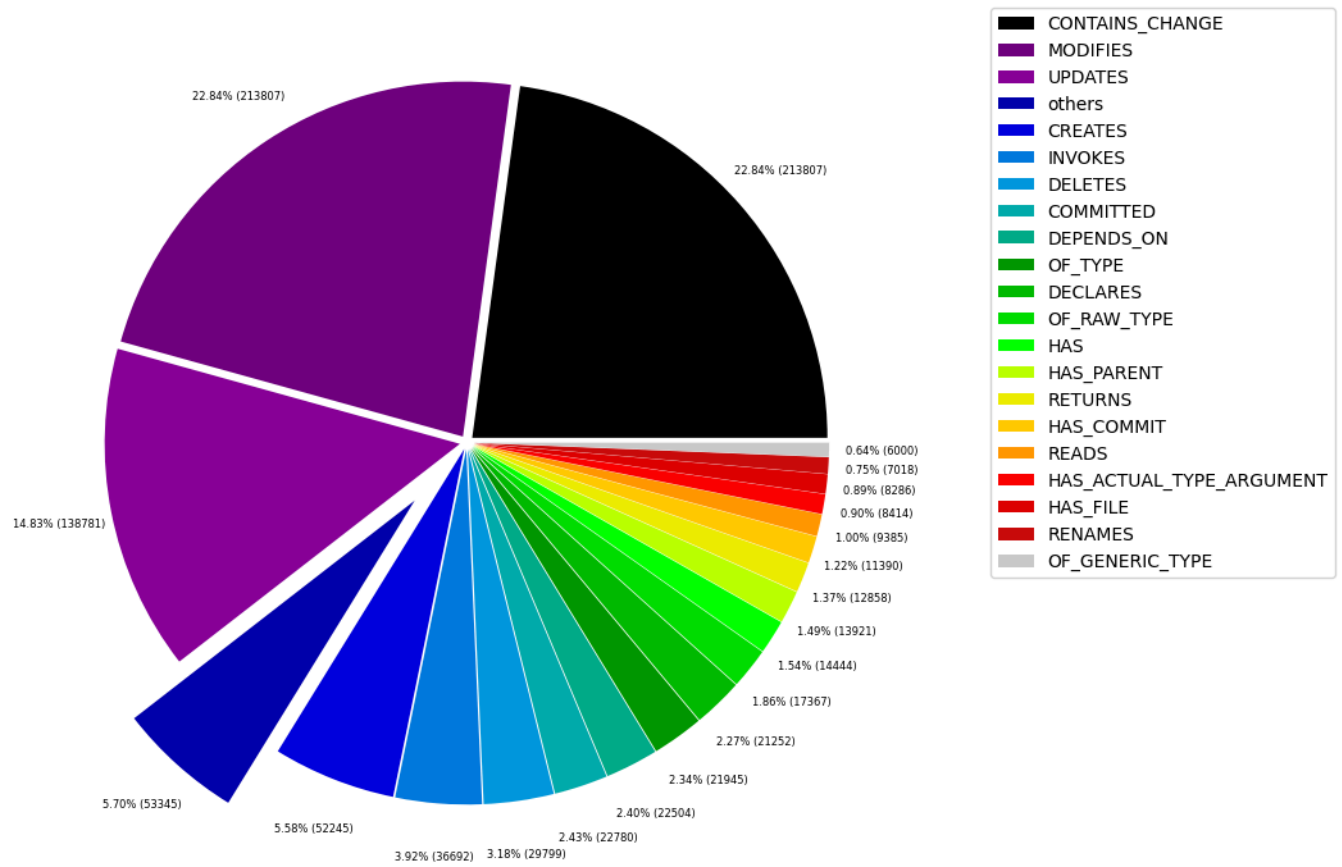


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

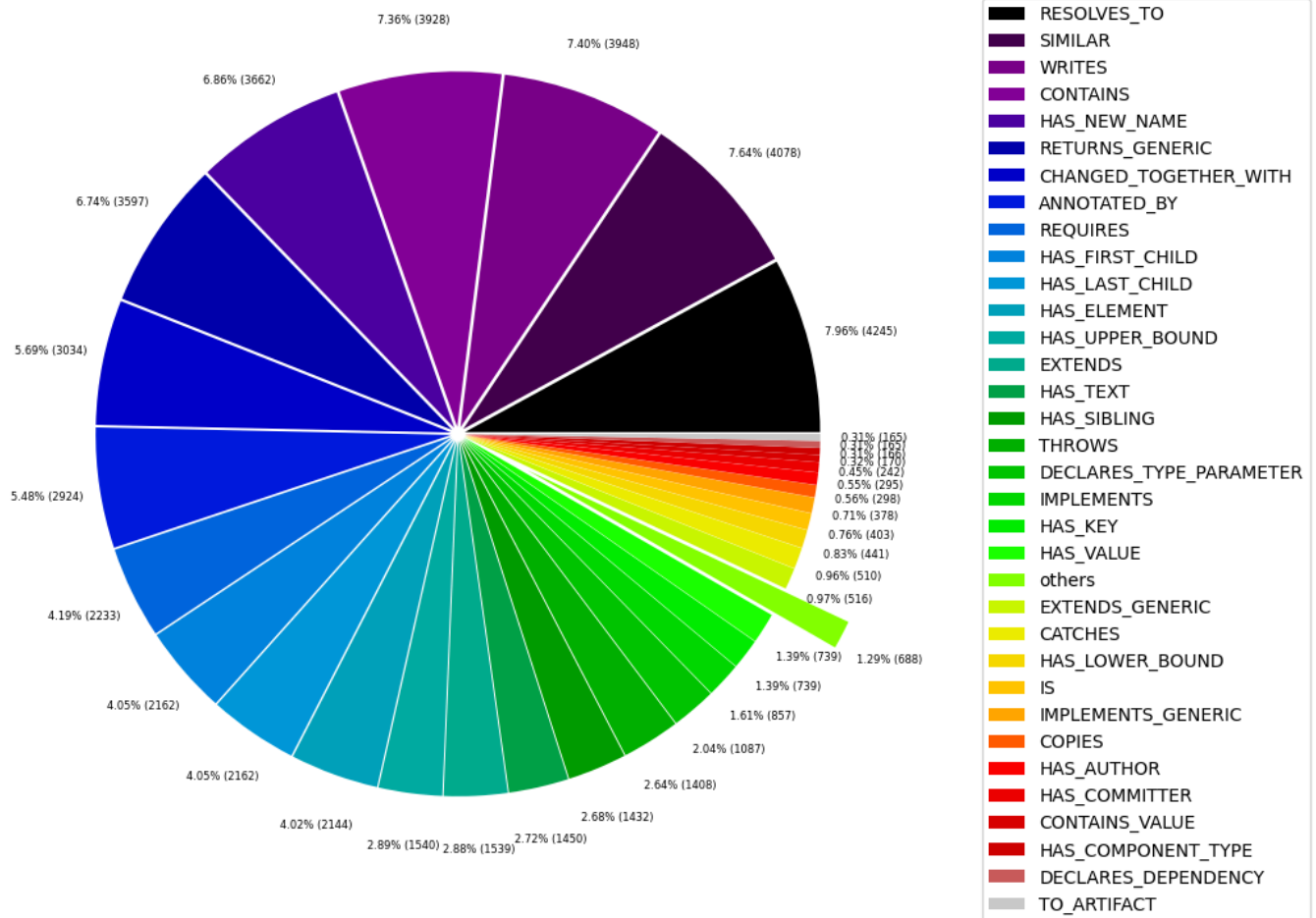
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	HAS_PROPERTY	1	0.000107
1	HAS_BRANCH	1	0.000107
2	HAS_HEAD	2	0.000214
3	THROWS_GENERIC	5	0.000534
4	EXCLUDES	5	0.000534
5	DESCRIBES	9	0.000961
6	HAS_ROOT_ELEMENT	11	0.001175
7	HAS_GOAL	16	0.001709
8	HAS_EXECUTION	16	0.001709
9	OF_NAMESPACE	18	0.001923
10	HAS_ATTRIBUTE	18	0.001923
11	INCLUDES_CONCEPT	19	0.002030
12	USES_PLUGIN	21	0.002243
13	IS_ARTIFACT	21	0.002243
14	HAS_CONFIGURATION	21	0.002243
15	REQUIRES_TYPE_PARAMETER	24	0.002564
16	REQUIRES_CONCEPT	28	0.002991
17	DECLARES_NAMESPACE	36	0.003846
18	HAS_DEFAULT	39	0.004166
19	COPY_OF	119	0.012713
20	HAS_TAG	129	0.013781
21	ON_COMMIT	129	0.013781
22	DECLARES_DEPENDENCY	165	0.017627
23	TO_ARTIFACT	165	0.017627
24	HAS_COMPONENT_TYPE	166	0.017734
25	CONTAINS_VALUE	170	0.018162
26	HAS_COMMITTER	242	0.025854
27	HAS_AUTHOR	295	0.031516
28	COPIES	298	0.031836
29	IMPLEMENTS_GENERIC	378	0.040383

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

<Figure size 640x480 with 0 Axes>

Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numbe
0	[Git, Update, Change]	UPDATES	[File, Git]	138781	138781	
1	[Git, Update, Change]	MODIFIES	[File, Git]	138781	138781	
2	[Git, Commit]	CONTAINS_CHANGE	[Git, Update, Change]	138781	11390	
3	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	44929	11390	
4	[Git, Change, Create]	CREATES	[File, Git]	44929	44929	
5	[Git, Change, Create]	MODIFIES	[File, Git]	44929	44929	
6	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	22781	11390	
7	[Git, Change, Delete]	DELETES	[File, Git]	22781	22781	
8	[Git, Change, Delete]	MODIFIES	[File, Git]	22781	22781	
9	[Java, ByteCode, Method, Member]	INVOKES	[Java, ByteCode, Method, Member]	22341	13427	
10	[Git, Commit]	HAS_PARENT	[Git, Commit]	13912	11390	
11	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	11390	1	
12	[Author, Git, Person]	COMMITTED	[Git, Commit]	11390	295	
13	[Committer, Git, Person]	COMMITTED	[Git, Commit]	11390	242	
14	[Java, ByteCode, Method, Member]	HAS	[Java, ByteCode, Parameter]	8517	13427	
15	[Java, ByteCode, Method, Member]	READS	[Java, ByteCode, Member, Field]	8430	13427	
16	[Repository, File, Git]	HAS_FILE	[File, Git]	8286	1	
17	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	7018	11390	
18	[Git, Change, Rename]	MODIFIES	[File, Git]	7018	7018	
19	[Git, Change, Rename]	CREATES	[File, Git]	7018	7018	
20	[Git, Change, Rename]	RENAMES	[File, Git]	7018	7018	
21	[Git, Change, Rename]	DELETES	[File, Git]	7018	7018	
22	[Java, ByteCode, Parameter]	OF_TYPE	[Type, File, Java, ByteCode, javaType]	6160	13306	
23	[File, Git]	HAS_NEW_NAME	[File, Git]	3662	8286	
24	[Java, ByteCode, Bound]	OF_RAW_TYPE	[Type, File, Java, ByteCode, javaType]	3466	7245	
25	[Java, ByteCode, Bound, ParameterizedType]	OF_RAW_TYPE	[Type, File, Java, ByteCode, javaType]	3114	7286	
26	[Java, ByteCode, Bound, ParameterizedType]	HAS_ACTUAL_TYPE_ARGUMENT	[Java, ByteCode, Bound, WildcardType]	2950	7286	
27	[Java, ByteCode, Parameter]	OF_GENERIC_TYPE	[Java, ByteCode, Bound, ParameterizedType]	2698	13306	
28	[Java, ByteCode, Constructor, Method, Member]	WRITES	[Java, ByteCode, Member, Field]	2520	2111	
29	[Java, ByteCode, Bound, ParameterizedType]	HAS_ACTUAL_TYPE_ARGUMENT	[Java, ByteCode, TypeVariable, Bound]	2475	7286	

Graph Density

```
total_number_of_nodes (vertices): 301218
total_number_of_relationships (edges): 936040
-> total directed graph density: 1.0316538625191052e-05
-> total directed graph density in percent: 0.0010316538625191052
```